

Internship Report

Identification and Bioinformatic Analysis of Ion Channels That May Modulate Immunity and/or Extracellular Matrix in PDAC Stroma

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) is a highly desmoplastic cancer in which tumor cells coexist with a dense and heterogeneous stromal compartment. Because fibroblast states are linked to extracellular matrix remodeling, tissue architecture, and immune signaling, ion channels expressed in this compartment may contribute to biologically relevant programs beyond tumor cells alone. Here, a multi-level bioinformatic workflow was used to identify and interpret potassium channel candidates associated with PDAC stroma. The first analytical block relied on a stromal single-cell RNA sequencing (scRNA-seq) dataset to prioritize KCN genes across complementary cell-level and pseudobulk differential expression strategies. This step highlighted *KCNMA1* as a candidate for an activated contractile stromal pole and *KCNJ8* as a candidate for a more quiescent or resident pole. No equally robust leader emerged for the inflammatory cancer-associated fibroblast (iCAF) axis. A second validation step used a broader PDAC single-nucleus RNA sequencing (snRNA-seq) atlas, referred to here as the USER dataset, to reinsert these candidates into the full tumor microenvironment. This step showed that *KCNN4* behaves as a strong tumor epithelial counterpoint, whereas *KCNMA1* and *KCNJ8* remain on the fibroblast side. A local bulk RNA-seq model of BKC_a silencing in cancer-associated fibroblasts (CAFs) showed that *KCNMA1* knockdown modifies the transcriptome and is linked to lower myCAF and iCAF scores. Finally, spatial transcriptomic analyses placed *KCNMA1*, *KCNJ8*, and *KCNN4* back into tissue context and reinforced the preferential links of *KCNMA1* with CAF territories and of *KCNN4* with tumor epithelial territories. Overall, I propose here a simple stromal model structured around *KCNMA1*, *KCNJ8*, and *KCNN4*.

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List of Abbreviations and Definitions

BH Benjamini-Hochberg.

BKCa Large-conductance calcium-activated potassium channel complex.

CAF Cancer-associated fibroblast.

csCAF Complement-secreting cancer-associated fibroblast.

DEG Differentially expressed gene.

ECM Extracellular matrix.

EMT Epithelial-mesenchymal transition.

FDR False discovery rate.

FFPE Formalin-fixed paraffin-embedded.

GSEA Gene set enrichment analysis.

iCAF Inflammatory cancer-associated fibroblast.

ifCAF Interferon-associated cancer-associated fibroblast.

KCN Potassium channel gene family.

log2FC Log₂ fold change.

myCAF Myofibroblastic cancer-associated fibroblast.

PCA Principal component analysis.

PDAC Pancreatic ductal adenocarcinoma.

PSC Pancreatic stellate cell.

Pseudobulk Aggregation of single-cell or single-nucleus counts by biological sample to perform sample-level statistics.

qPSC Quiescent pancreatic stellate cell.

RNA-seq RNA sequencing.

scRNA-seq Single-cell RNA sequencing.

shBKCa Short-hairpin RNA-mediated BKCa silencing.

smPSC Smooth muscle-like pancreatic stellate cell.

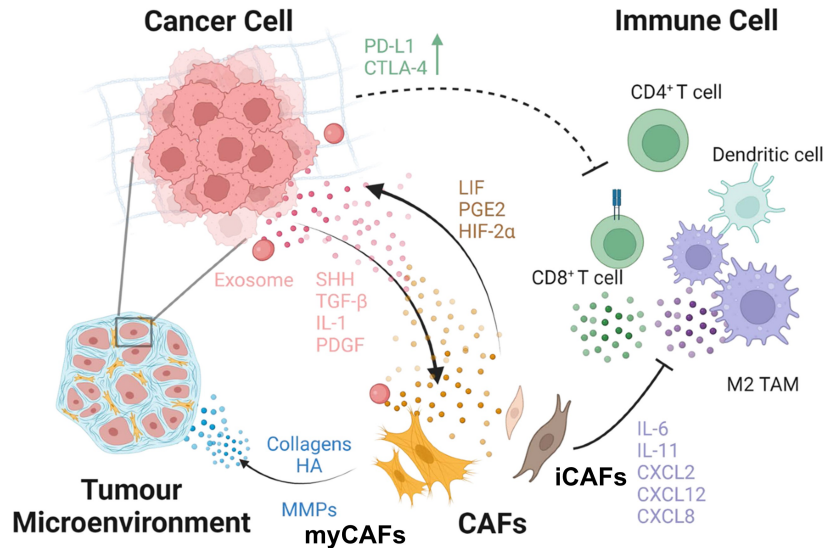
snRNA-seq Single-nucleus RNA sequencing.

ssGSEA Single-sample gene set enrichment analysis.

UMAP Uniform Manifold Approximation and Projection.

USER Project name used here for the Hwang et al. PDAC snRNA-seq validation atlas.

1. Introduction



Zhang, T., Ren, Y., Yang, P. *et al.* Cancer-associated fibroblasts in pancreatic ductal adenocarcinoma. *Cell Death Dis* **13**, 897 (2022). <https://doi.org/10.1038/s41419-022-05351-1>

Figure 1: Biological context of the PDAC stromal microenvironment. PDAC epithelial cells are embedded in a dense stromal microenvironment structured by fibroblast heterogeneity, extracellular matrix remodeling, and immune interactions. The scheme shows the myCAF/iCAF framework.

Pancreatic ductal adenocarcinoma (PDAC) is not only a disease of malignant epithelial cells. It is also a disease of tissue organization. Pancreatic cancer remains one of the deadliest cancers, with mortality remaining close to incidence worldwide and 5-year survival still around 13% in population-based statistics (Bray et al., 2024; SEER, 2026). One of its main features is a strong desmoplastic reaction in which tumor cells coexist with abundant extracellular matrix (ECM), immune cells, and heterogeneous fibroblasts. This stromal compartment, meaning the non-malignant supportive tissue surrounding tumor cells, affects tumor growth, limits drug penetration, and shapes local immune regulation and immune suppression (Ohlund et al., 2017; Elyada et al., 2019; Oh et al., 2023). For this reason, understanding stromal biology is important for understanding PDAC itself.

Cancer-associated fibroblasts (CAFs) are now recognized as a heterogeneous population rather than a single cell type. In PDAC, myofibroblastic CAFs (myCAFs) are usually described as matrix-rich and contractile, whereas inflammatory CAFs (iCAFs) are characterized by a more secretory and pro-inflammatory program, notably including cytokine production (Ohlund et al., 2017; Elyada et al., 2019). Recent single-cell studies also support the existence of more quiescent or resident stellate states and additional inflammatory or interferon fibroblast programs (Hosein et al., 2019; Oh et al., 2023). Here, this biological complexity is simplified into a practical contrast between an activated contractile stromal pole and a more quiescent resident stromal pole.

Previous work from the laboratory supports the idea that ion channels are not only passive

markers in PDAC, but can actively participate in tumor/stroma communication. In particular, SK2 was shown to respond to CAF signals and to reinforce signaling pathways associated with invasiveness and metastasis (Rapetti-Mauss et al., 2023). More broadly, ion channels are increasingly recognized as active contributors to cancer biology, to the point that cancer hallmarks have been discussed in the framework of oncochannelopathies (Prevarskaya et al., 2018).

Ion channels are therefore relevant in this context because they regulate membrane potential, calcium entry, secretion, cell volume, and mechanical responses. In fibroblasts, these functions can influence contraction, matrix deposition, and communication with neighboring cells. However, ion channels are often expressed at relatively low transcript levels, especially in sparse single-cell or spatial assays, which makes them easy to miss if analysis relies only on transcript abundance alone. This does not mean that they are biologically unimportant, because even a limited number of ion channels can control resting membrane potential and thereby influence membrane transport, intracellular signaling, and cellular metabolism. These effects can translate into major functional changes (Scruggs et al., 2020; Jiang et al., 2019). In this project, I focused on the potassium channel (KCN) family because it is the largest ion channel family and because potassium channel biology is a clear area of interest in the host laboratory.

The broader project initially identified a union of 25 KCN candidates across the stromal discovery analyses. Because the report is limited in size and number of figures, I focus on three genes that best structure the story. *KCNMA1*, which encodes the pore-forming alpha subunit of the BKCa channel, is used as the main activated stromal candidate. *KCNJ8* is used as a candidate for a more quiescent or resident stromal pole. *KCNN4* is used as a tumor epithelial counterpoint that helps distinguish stromal and epithelial territories. This interpretation of *KCNN4* as a tumor signal is also consistent with PDAC literature, where *KCNN4*-mediated signaling has already been linked to tumor progression (Jiang et al., 2019). Other KCN genes remain interesting and will be discussed briefly in the final schematic and perspectives.

2. Objectives

The main objective of this internship was to identify ion channels that may modulate extracellular matrix organization and/or stromal immune functions in PDAC. To address this question, the report followed four linked goals. First, I sought to prioritize stromal potassium channel candidates in a stromal scRNA-seq dataset. Second, the focus shifted to validating and repositioning these candidates in a broader PDAC snRNA-seq atlas containing both stromal and tumor compartments. Third, I set out to test whether BKCa silencing in CAFs is associated with coherent transcriptomic changes. Fourth, the final step involved placing the main candidates back into tissue space using spot-level spatial transcriptomics. The ultimate goal was to build a coherent and readable stromal interpretation supported by several data layers.

3. Materials and Methods

3.1. Overall Bioinformatic Workflow

This work is presented as a workflow rather than as a finalized pipeline. It combines four analytical blocks that match the main figures of the report: stromal single-cell prioritization, external validation in a broader snRNA-seq atlas, local bulk RNA-seq perturbation around BKCa signaling, and spatial transcriptomic localization. The workflow is currently being reorganized into a more reusable pipeline for later use in the project, in line with current practical single-cell analysis frameworks (Luecken and Theis, 2019; He et al., 2022).

The project relied mainly on **R** and **Python** for data processing, statistical analysis, and figure generation. The main R packages were **Seurat** (single-cell analysis framework), **DESeq2** (count-based differential expression for RNA-seq), **MAST** (single-cell hurdle-model differential expression), **fgsea** (fast gene set enrichment analysis), **msigdbR**, **escape**, **ggplot2**, **ggrepel**, **VennDiagram**, **pheatmap**, and standard data-handling packages such as **dplyr** and **tibble**. The main Python packages were **scanpy** (single-cell analysis toolkit), **pandas**, and the **Hotspot** framework. All analyses performed during this internship are available in the GitHub repository `ym22000/pdac-ion-channels-m1`.

3.2. Moffitt Stromal scRNA-seq Discovery Block

The first analytical block used a stromal Seurat object derived from the coordinated single-cell PDAC resource from Oh et al. (Oh et al., 2023). ScRNA-seq measures the transcriptome of individual whole cells after dissociation. The full atlas contains several compartments, including tumor, immune, endothelial, and stromal populations, but the report starts from the stromal object because the focus is stromal. A broader tumor/stroma comparison was addressed later with the USER atlas, which is more suitable for that purpose.

The stromal object contained 21,469 stromal cells from 40 samples and 29 patients, distributed across eight stromal cell types. UMAP plots were read from the Seurat object and are based on normalized single-cell expression values. The object also contained the original activated-stroma and normal-stroma scores from the Moffitt framework. I used the difference between these two scores to define a broad **StromaState** variable. This broad two-state view was kept because it captures the original Moffitt activated-versus-normal stromal axis and provides a complementary framework to the finer subtype-specific contrasts.

Four complementary differential expression strategies were combined in Fig. 2b. Test 1 was a patient-aware pseudobulk **DESeq2** comparison between activated and more quiescent stromal states. In this test, raw counts were aggregated by patient and state before running a count-based negative binomial model with a Wald test (Love et al., 2014). Test 2 was a patient-aware pseudobulk **DESeq2** subtype/rest comparison, again using aggregated raw counts. Test 3 was a cell-level hurdle model designed to separate gene detection from positive-expression behavior. Such models are useful in zero-inflated data because a gene can differ either by being detected

more often or by being more strongly expressed when detected (Finak et al., 2015). Test 4 was a standard MAST analysis using log-normalized single-cell expression values (Finak et al., 2015). The Venn diagram therefore summarizes convergence across different input matrices and different statistical assumptions, which is important because single-cell results can vary substantially with analysis choices (Vieth et al., 2019).

For the direct qPSC/myCAF comparison shown in Fig. 2d, raw counts were aggregated into patient-aware pseudobulk samples. Only paired patients with enough cells in both subtypes were retained, leading to 29 paired patients, 5,386 qPSC cells, and 4,699 myCAF cells. DESeq2 was run with the design `~ patient + subtype`. Hallmark gene set enrichment analysis (GSEA) was then performed with `fgsea` using the Wald statistic ranking (Subramanian et al., 2005; Korotkevich et al., 2021).

3.3. USER snRNA-seq Validation Block

The second analytical block used a broader PDAC single-nucleus RNA sequencing (snRNA-seq) dataset from Hwang et al. (Hwang et al., 2020). snRNA-seq measures nuclear RNA instead of whole-cell RNA and is often more stable in fibrotic tissues. Here, this dataset is called the USER dataset. The key advantage of this dataset is that it contains the full microenvironment rather than fibroblasts alone, which makes it useful for placing stromal candidates back into a tumor/stroma context.

The global USER figure was built from published metadata and UMAP coordinates. The dataset contained 88,031 nuclei, including 17,537 fibroblast nuclei. The proportions of tumor and fibroblast nuclei varied noticeably across patients. This heterogeneity needed to be made explicit before performing the tumor/fibroblast comparison shown in Fig. 3b.

So for the tumor/fibroblast contrast, raw counts were aggregated by patient and cell subset. Only patients containing both tumor and fibroblast nuclei were kept. Genes with very low total counts were removed before analysis. DESeq2 was then run with the design `~ patient + group`. The final contrast included 15 paired patients, 43,817 tumor nuclei, 17,537 fibroblast nuclei, and 18,993 tested genes. Also, differential expression analyses were performed across the broader set of major USER cell subsets, but are not detailed here because the tumor/fibroblast contrast is the most directly informative for repositioning stromal KCN candidates.

3.4. Bulk RNA-seq Analysis of BKCa Silencing in CAFs

The third analytical block used a local bulk RNA-seq dataset generated in the host laboratory. In this experiment, BKCa was silenced through short-hairpin RNA targeting *KCNMA1* in cultured human CAFs, creating a **shBKCa** condition compared with non-targeting controls. Here, BKCa refers to the large-conductance calcium-activated potassium channel complex built around the pore-forming *KCNMA1* gene and the auxiliary beta-subunit genes *KCNMB1* and *KCNMB4*. Bulk RNA-seq measures the average transcriptome of each sample rather than individual cells. The comparison included four control samples and four **shBKCa** samples, and gene-level counts

were collapsed into 17,889 genes.

Principal component analysis (PCA) was used as an internal quality control step on normalized transformed expression values to confirm separation of replicates by condition, although this PCA is not shown here. Differential expression was performed on raw count data with **DESeq2** (Love et al., 2014). Hallmark pathway enrichment was performed with **fgsea**, using the gene-level log2FC ranking and following the general GSEA framework (Subramanian et al., 2005; Korotkevich et al., 2021). CAF signature heatmaps were built from normalized expression values scaled by gene across samples, corresponding in the legend to a row z-score. At signature level, condition differences were summarized with Wilcoxon rank-sum tests. The myCAF and iCAF signatures were based on published PDAC CAF states from Ohlund et al. and Elyada et al., and an additional ifCAF signature was included to capture an interferon-associated fibroblast program in the bulk perturbation model (Ohlund et al., 2017; Elyada et al., 2019).

3.5. Spatial Transcriptomics and Signature Projection

The spatial block used a Visium FFPE PDAC dataset from Khaliq et al. (Khaliq et al., 2024). Spatial transcriptomics measures RNA expression at spot level while preserving tissue coordinates. In this dataset, the 30 tissue sections represented 91,496 spots in total, including 3 normal pancreas sections (9,820 spots), 10 PDAC sections (35,458 spots), 12 liver metastasis sections (28,520 spots), and 5 lymph node sections (17,698 spots). Structural annotations and multicellular ecotypes from the reference study were reused as a tissue framework.

I focused mainly on normal pancreas and primary PDAC sections because the focus concerns pancreatic stromal organization. Liver metastasis and lymph node sections remain useful in the larger project, but they are not necessary to support the argument developed here.

Spatial interpretation relied on two complementary levels. First, local Hotspot maps were used to visualize spatially coherent territories for *KCNMA1* and *KCNN4*. **Hotspot** framework detects locally structured patterns rather than isolated high spots, which makes it useful for distinguishing real tissue territories from scattered signal (DeTomaso and Yosef, 2021). A 30-nearest-neighbor graph was used to capture local spatial structure while avoiding excessive smoothing across broader tissue regions. A Bernoulli model was used because many KCN genes were sparsely detected across spots, making presence/absence patterns more informative than raw count magnitude alone. Second, reference signatures were projected at spot level and compared with KCN signals using Spearman rank correlation. Spearman correlation was preferred because spot-level values are sparse, non-gaussian, and better interpreted through monotonic association than strict linearity.

The tumor epithelial signature used in Fig. 6a was defined in the laboratory from epithelial markers considered informative in this project (*EPCAM*, *MUC1*, *KRT7*, *KRT8* and *KRT18*). The CAF signature used in Fig. 6b came from the reference spatial study and was projected as a paper-derived ssGSEA-like score (Khaliq et al., 2024). In practice, the spatial analysis combined R and Python processing using **Seurat**, **escape**, **scanpy**, **pandas**, and **Hotspot**.

3.6. Statistics and Multiple Testing

Effect size is mainly described here with $\log_2\text{FC}$, and statistical significance is summarized with adjusted p-values or false discovery rate (FDR). In **DESeq2**, p-values came from Wald tests on count-based negative binomial models (Love et al., 2014). In **MAST**, p-values came from a hurdle framework applied to log-normalized expression (Finak et al., 2015). Because omics analyses test many genes and pathways at the same time, multiple testing correction was performed with the Benjamini-Hochberg (BH) method rather than Bonferroni correction. BH was preferred because it controls the expected proportion of false discoveries while preserving reasonable sensitivity in exploratory transcriptomic analyses.

Single-cell and single-nucleus differential results were interpreted together with pseudobulk analyses to avoid over-reliance on cell-level pseudoreplication. Wilcoxon tests were used for non-parametric comparison of summary scores, such as CAF signatures in bulk RNA-seq. Spearman correlation was used in the spatial block for spot-level association analyses.

4. Results and Discussion

4.1. A Stromal KCN Core Was Identified in the Moffitt Dataset

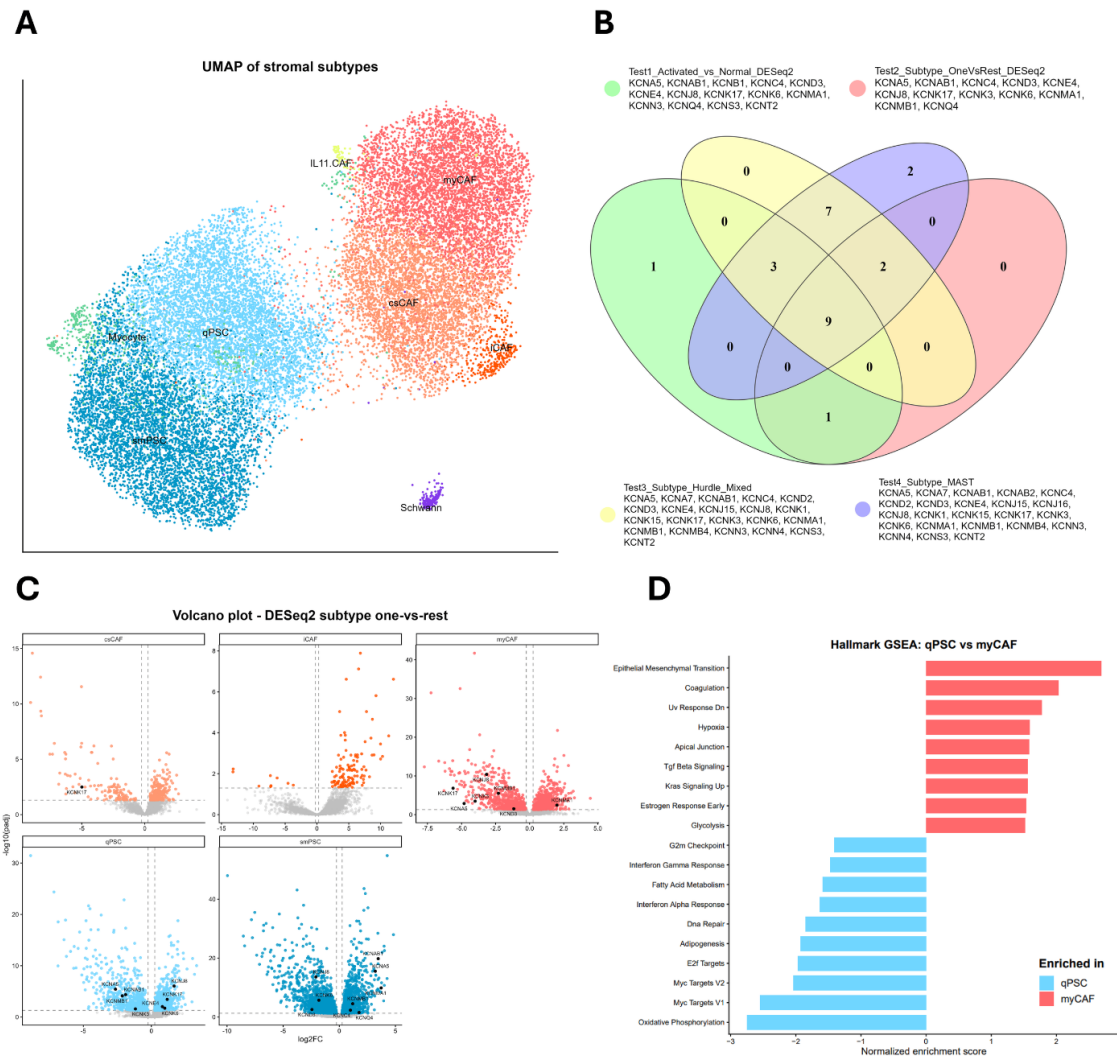


Figure 2: Stromal single-cell prioritization in the Moffitt dataset. **a** UMAP representation of stromal cell subtypes. **b** Four-test Venn summary showing overlap between complementary differential expression strategies used to prioritize KCN candidates across cell-level and pseudobulk approaches. **c** Volcano plots from the pseudobulk subtype/rest analysis highlighting KCN behavior across stromal subtypes. **d** Hallmark GSEA comparison between qPSC and myCAF.

In Fig. 2a, the stromal compartment is not a homogeneous fibroblast cloud. It is organized into qPSC, smPSC, myCAF, csCAF, and iCAF states, together with smaller associated populations. This matters because the project question is not only whether a KCN gene is stromal, but also which stromal state it follows. The stromal object contained 21,469 cells in total, including 5,743 qPSCs, 5,521 smPSCs, 4,809 myCAFs, 4,242 csCAFs, and only 338 iCAFs.

Figure 2b shows that the four differential expression strategies converge on a compact robust core. The full union of candidates was useful to preserve potentially relevant KCN genes at an exploratory stage, but the strict intersection is the most robust part of the result. Nine KCN

genes were detected by all four methods (*KCNA5*, *KCNAB1*, *KCNC4*, *KCND3*, *KCNE4*, *KCNJ8*, *KCNK17*, *KCNK6*, and *KCNMA1*). This convergence is important because the four methods do not test exactly the same thing. The two pseudobulk tests are conservative and sample-aware. The hurdle-based single-cell tests (Test 3 and 4) are more sensitive to sparse detection behavior. The **StromaState** comparison was also worth keeping because it reconnects the subtype-based analysis to the original activated/normal stromal logic from the Moffitt framework.

In Fig. 2c, the subtype-specific effect sizes refine this convergence. In the pseudobulk subtype/rest analysis, *KCNMA1* was strongly enriched in smPSC with $\log_2\text{FC} = 3.73$ ($\text{FDR} = 1.30 \times 10^{-10}$) and was also enriched in myCAF with $\log_2\text{FC} = 2.00$ ($\text{FDR} = 3.63 \times 10^{-3}$). In contrast, *KCNJ8* was enriched in qPSC with $\log_2\text{FC} = 1.67$ ($\text{FDR} = 9.11 \times 10^{-7}$) and strongly depleted in myCAF with $\log_2\text{FC} = -3.16$ ($\text{FDR} = 4.04 \times 10^{-11}$). *KCNN4* did not emerge as a significant stromal subtype marker in this stromal setting. These values align with the primary stromal polarity established here, wherein *KCNMA1* delineates an activated, contractile pole, whereas *KCNJ8* characterizes a more quiescent, resident phenotype.

Figure 2d shows that this polarity is not limited to single genes. The myCAF side is enriched in epithelial-mesenchymal transition, TGF- β signaling, coagulation, hypoxia, glycolysis, and KRAS signaling up. In contrast, qPSC is enriched in oxidative phosphorylation, fatty acid metabolism, interferon response, and DNA repair. The iCAF axis was explored, but it remained less robust in this dataset. This is consistent with the low number of iCAF cells. For example, *KCNJ8* showed $\log_2\text{FC} = -3.34$ in iCAF but remained non-significant after correction ($\text{FDR} = 0.311$), and *KCNMA1* showed a modest positive signal without significance. Here, iCAF is therefore treated as an explored but unresolved axis rather than as a fourth strong KCN-centered pole.

4.2. USER Validation Repositioned the Candidates Along a Tumor–Stroma Axis

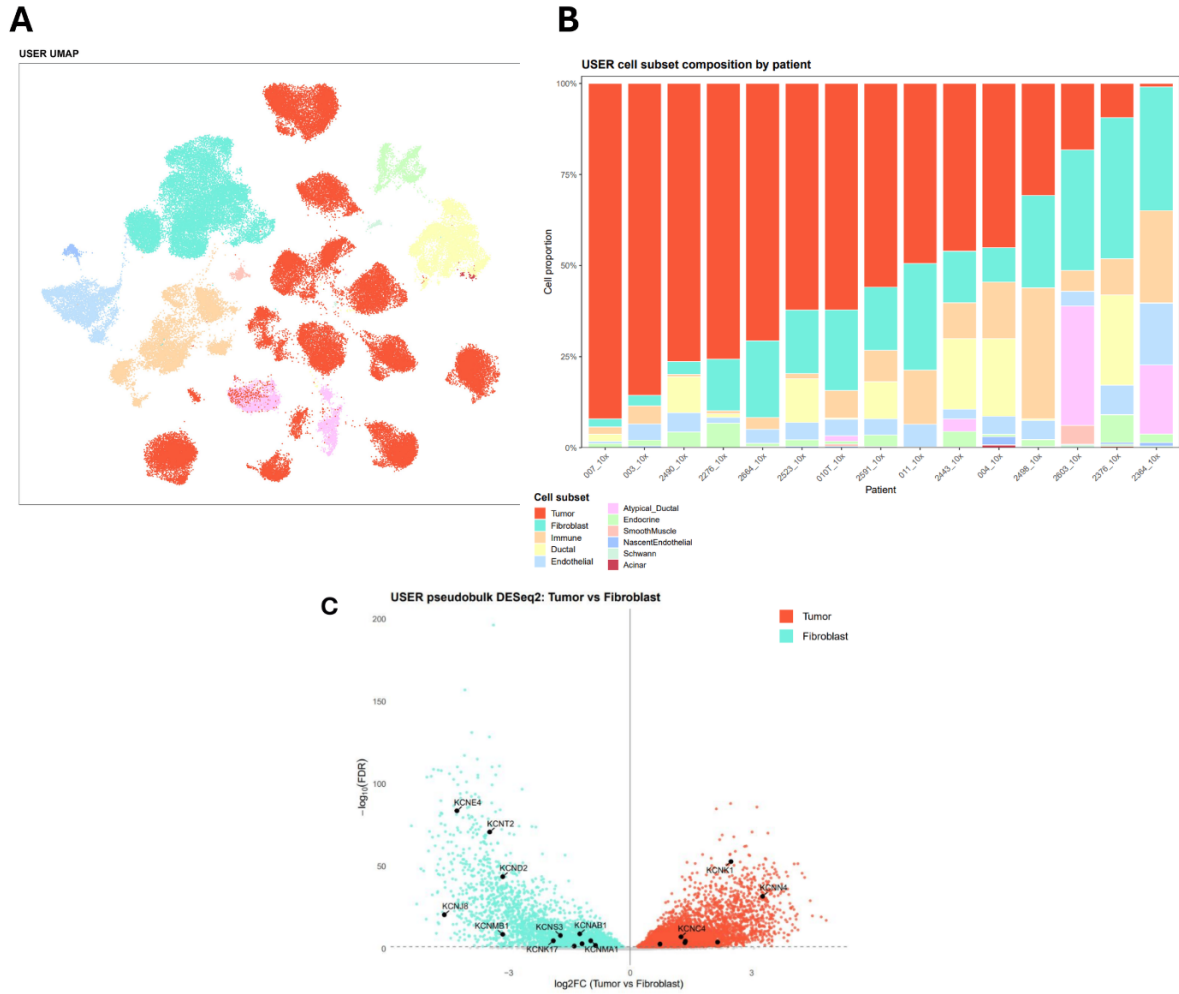


Figure 3: External validation in the USER snRNA-seq dataset. **a** UMAP of the USER PDAC atlas used to reinsert stromal candidates into a broader tumor microenvironment context. **b** Cell-type composition across patients, shown to highlight inter-patient variability and to justify patient-aware pseudobulk analyses. **c** Volcano plot of the pseudobulk Tumor/Fibroblast comparison, used to position the selected KCN candidates along a stromal/tumor axis.

In Fig. 3a, the candidates are repositioned in a broader whole-tissue context. This is important because a gene that looks stromal inside a fibroblast object can still be shared with other compartments when the whole tumor microenvironment is considered. The USER atlas is particularly useful for this because it contains tumor, fibroblast, immune, endothelial, and ductal compartments in the same framework.

Figure 3b shows strong inter-patient variation in cellular composition. Some samples are largely tumor-dominated, whereas others contain larger fibroblast fractions. This justifies the patient-aware pseudobulk design used for the tumor/fibroblast contrast. Without this design, large patient-specific composition effects could be mistaken for true gene-specific biology.

A volcano plot summarizes differential expression by showing effect size on the x-axis and statistical significance on the y-axis. In Fig. 3c, the simplest global question is whether a gene is closer

to tumor or fibroblast in the full atlas. In this direct Tumor/Fibroblast comparison, *KCNN4* was strongly tumor-up, with $\log_2\text{FC} = 3.27$ and $\text{FDR} = 1.23 \times 10^{-32}$. In contrast, *KCNJ8* was strongly fibroblast-up, with $\log_2\text{FC} = -4.59$ and $\text{FDR} = 1.92 \times 10^{-21}$. *KCNMA1* also stayed on the fibroblast side, with $\log_2\text{FC} = -0.975$ and $\text{FDR} = 1.13 \times 10^{-5}$, although the effect size was smaller than for *KCNJ8*. This result is informative for the general model. It confirms *KCNN4* as a tumor epithelial counterpoint, keeps *KCNJ8* as a strongly fibroblast-enriched gene, and places *KCNMA1* in a more stromal and microenvironmental position. This validation is also coherent with the Moffitt prioritization, because *KCNJ8* and *KCNMA1* belonged to the robust 4/4 core, whereas *KCNN4* emerged in only 2 of the 4 DEG strategies.

4.3. BKCa Knockdown in CAFs Altered Programs Linked to Activated Stroma

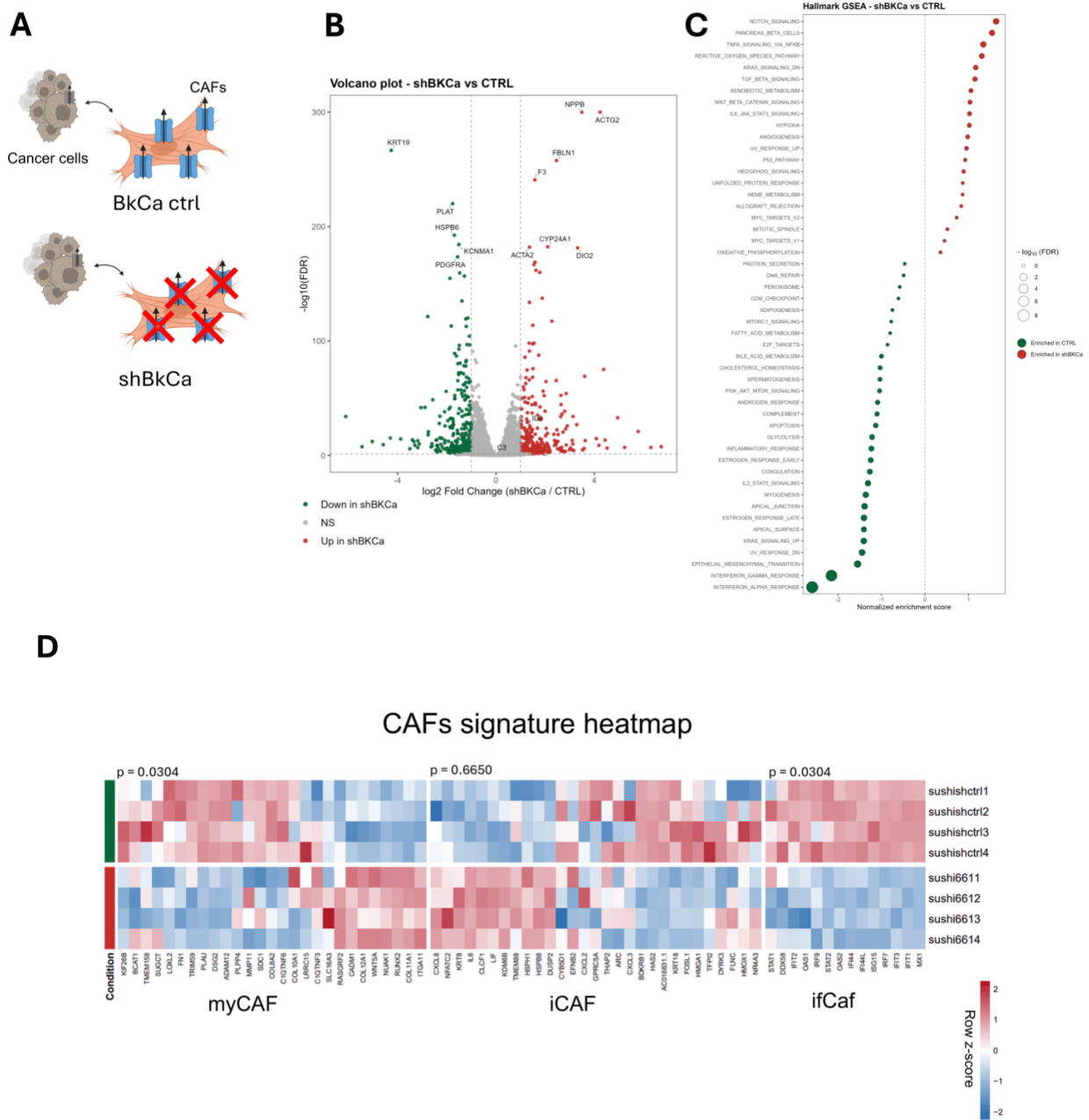


Figure 4: Bulk RNA-seq perturbation analysis in a BKCa CAF model. **a** Experimental overview of the control and shBKCa comparison used in the local bulk RNA-seq block. **b** Volcano plot of differential expression between shBKCa and control samples. **c** Hallmark GSEA summary used to identify broad transcriptomic programs associated with the perturbation. **d** Heatmap of selected CAF signatures, including myCAF, iCAF, and ifCAF markers, used to examine whether the perturbation is associated with stromal program shifts.

In Fig. 4a, the question is no longer where a gene is expressed, but whether BKCa silencing in shBKCa CAFs changes the transcriptome relative to control CAFs. This part adds a functional layer from the host laboratory to the descriptive analyses.

In Fig. 4b, the perturbation worked. *KCNMA1* itself was reduced in shBKCa, with $\log_2\text{FC} =$

-1.51 and an extremely low adjusted p-value. The volcano plot shows changes in structural or stromal genes such as *ACTA2*, *ACTG2*, *FBLN1*, and *PDGFRA*, reinforcing the idea that BKC_a signaling is linked to stromal cell state.

At pathway level, Fig. 4c shows that the strongest significant enrichments are found on the control side, especially interferon alpha response, interferon gamma response, and epithelial-mesenchymal transition. In other words, the major effect of BKC_a silencing in this model is not the appearance of one dominant new program, but rather the weakening of already structured programs.

In Fig. 4d, the CAF heatmap is useful for the stromal interpretation because it suggests a global shift in CAF state rather than a single binary marker change. The clearest pattern is seen for the iCAF signature, which is clearly lower in **shBKC_a** than in controls (mean score 0.84 in CTRL / -0.84 in **shBKC_a**; nominal Wilcoxon $p = 0.0304$). The myCAF signature also tends to decrease in **shBKC_a** (mean score 0.12 in CTRL / -0.12 in **shBKC_a**; $p = 0.0304$), but the heatmap indicates more heterogeneity across samples than for iCAF. In contrast, the eCAF signature is more variable and does not show a robust directional shift ($p = 0.665$). Although limited by the small cohort size ($n = 8$), this pattern supports the idea that BKC_a is not only linked to one CAF marker set, but more broadly acts as a driver of CAF signatures and CAF plasticity.

4.4. Spatial Analyses Distinguished Stromal and Epithelial Territories

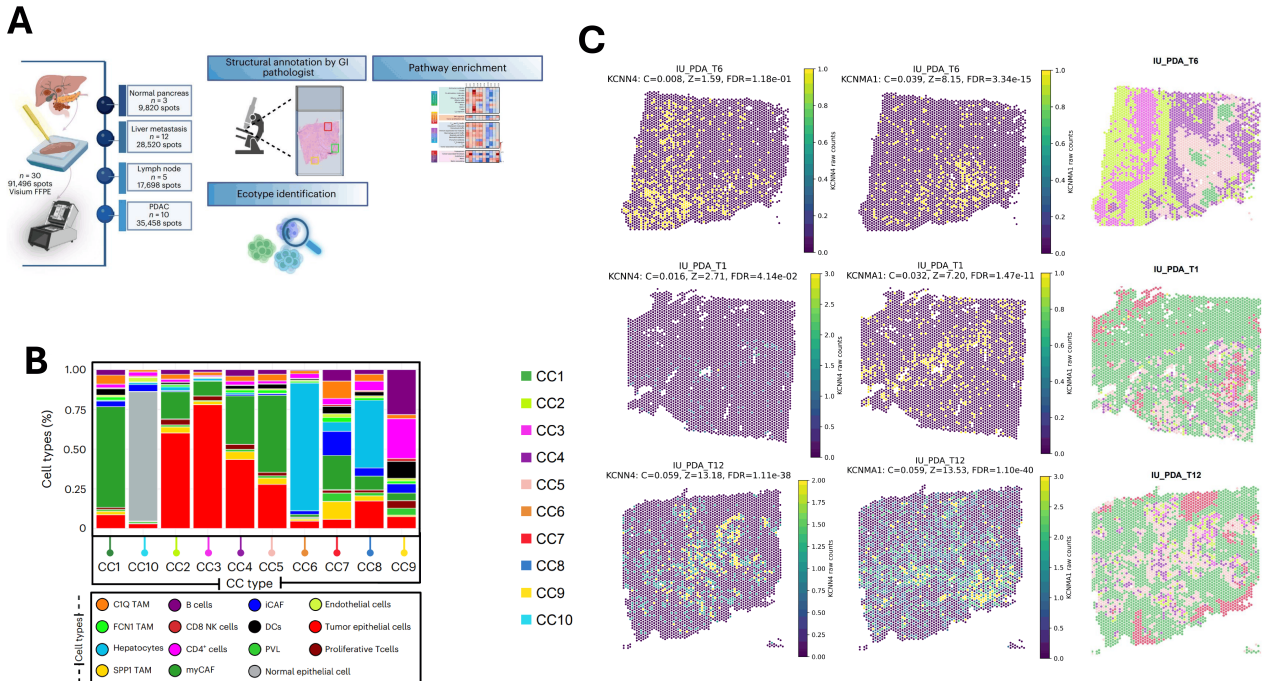


Figure 5: Spatial transcriptomic context for the main KCN candidates. **a** Overview of the spatial transcriptomic dataset, including section types, spot numbers, and ecotype information used. **b** Ecotype composition summary from the reference PDAC study. **c** Representative Hotspot maps for *KCNN4* and *KCNMA1* in selected PDAC sections (T6, T1, and T12), shown together with the neighboring ecotype context.

In Fig. 5a, the analysis returns to tissue architecture. For the Hotspot results discussed in Fig. 5, the Z score corresponds to the standardized strength of spatial autocorrelation for a gene within one section, C is the associated scaled autocorrelation coefficient, and FDR is the Benjamini-Hochberg-adjusted significance of this spatial test. Higher positive Z and C values indicate a more coherent patch spatial pattern rather than diffuse expression. As described in the methods, Hotspot here was run with a 30-nearest-neighbor graph and a Bernoulli model, which is appropriate for preserving local tissue structure while handling sparse spot-level detection. For the signature projections discussed in Fig. 6, ρ denotes the Spearman rank correlation between a KCN gene and the projected signature across spots in one section (DeTomaso and Yosef, 2021). The spatial dataset shown in Fig. 5a spans several anatomical contexts, but as said before, the report focuses mainly on normal pancreas and primary PDAC sections because these are relevant here. Fig. 5b recalls the multicellular ecotypes/niches described in the reference spatial study.

In Fig. 5c, *KCNMA1* is repeatedly detected in stromal or mixed microenvironmental territories (ecotypes CC1 and CC5 in the paper), whereas *KCNN4* is more variable and visually closer to epithelial tumor territories (ecotypes CC2 and CC3). A complementary spot level Wilcoxon analysis across primary PDAC sections supported this interpretation. Therefore, the ecotype maps next to these panels help interpret these signals as territory associations rather than pure single-cell identities, because Visium spots average transcripts from small multicellular neighborhoods.

Although it is not shown here, *KCNJ8* displayed a stronger spatial Hotspot signal in normal pancreas than in primary PDAC, with a median Z score of 5.17 across normal sections / 1.15 across PDAC sections, together with a higher median C value (0.0238 / 0.00617). This was also consistent with the raw spot counts, because *KCNJ8* was much more abundant in normal pancreas than in primary PDAC (mean raw count across all spots, 1.20 versus 0.10 counts per spot; 55.1% versus 9.1% positive spots). Overall, *KCNJ8* is therefore mainly supported by the stromal polarity seen in Moffitt and confirmed in USER, while its spatial pattern also suggests relative enrichment in non-diseased pancreatic tissue compared with PDAC.

4.5. Signature Projection Strengthened the Contrast Between *KCNMA1* and *KCNN4*

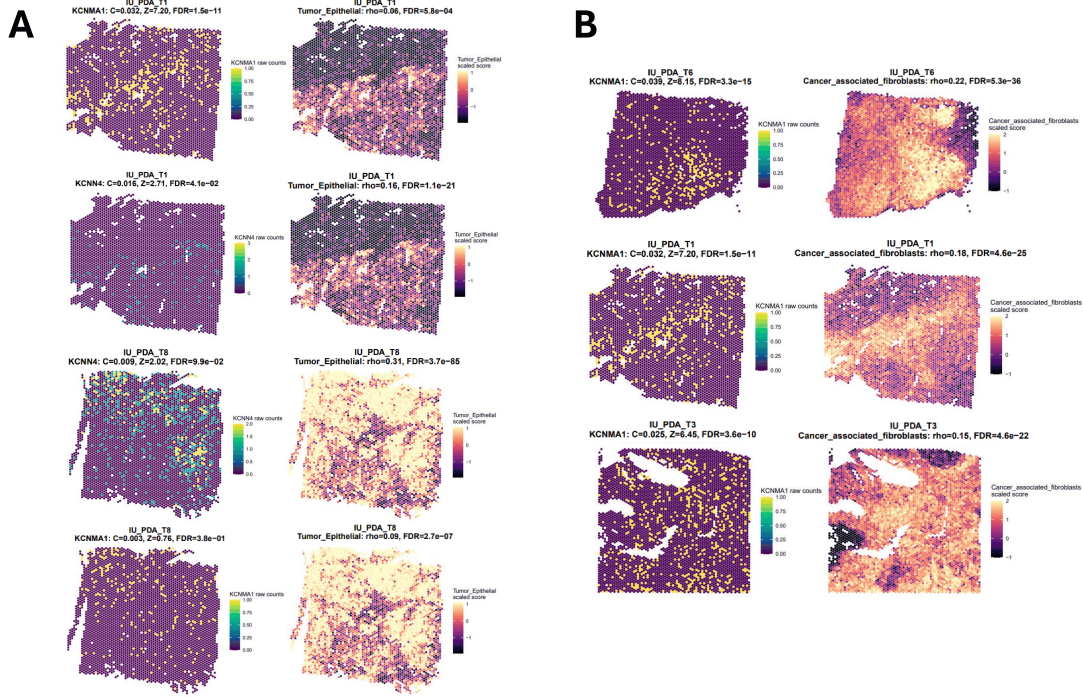


Figure 6: Spatial projection of tumor epithelial and CAF signatures. **a** Spot-level Spearman correlation projections comparing *KCNMA1* and *KCNN4* with a manually defined tumor epithelial signature in selected PDAC sections T1 and T8. **b** Spot-level comparison between *KCNMA1* and the cancer-associated fibroblast score in representative sections T6, T1, and T3.

In this section, the spatial reading is refined by comparing gene expression to continuous biological signatures. This is because a spot-level gene map alone can remain ambiguous, whereas a spot-level correlation with a known reference program gives a more quantitative interpretation. This analysis was also motivated by the observation that *KCNMA1* seemed to overlap with other spatial patterns that appeared more local than the ecotype structure.

In Fig. 6a, *KCNN4* follows the manually defined tumor epithelial signature more clearly than *KCNMA1*. In T1, the tumor epithelial score correlated with *KCNN4* at $\rho = 0.16$ ($\text{FDR} = 1.1 \times 10^{-21}$), whereas the correlation with *KCNMA1* was lower at $\rho = 0.06$ ($\text{FDR} = 5.8 \times 10^{-4}$). In T8, the same trend remained visible, with $\rho = 0.31$ for *KCNN4* ($\text{FDR} = 3.7 \times 10^{-85}$) and $\rho = 0.09$ for *KCNMA1* ($\text{FDR} = 2.7 \times 10^{-7}$). At the raw spot level, *KCNN4* was also nearly absent from normal pancreas but clearly more detectable in primary PDAC (mean 0.159 versus 0.0033 counts per spot; 12.8% versus 0.32% positive spots). These values are consistent with the USER tumor/fibroblast result and strengthen the interpretation of *KCNN4* as a tumor epithelial counterpoint.

In Fig. 6b, *KCNMA1* provides the opposite information. Across the sections shown, *KCNMA1* positively tracked the CAF score, with $\rho = 0.22$ in T6 ($\text{FDR} = 5.3 \times 10^{-36}$), $\rho = 0.18$ in T1 ($\text{FDR} = 4.6 \times 10^{-25}$), and $\rho = 0.15$ in T3 ($\text{FDR} = 4.6 \times 10^{-22}$). At the raw spot level, *KCNMA1* was also more prevalent in primary PDAC than in normal pancreas (mean 0.234 versus 0.158

counts per spot; 17.4% versus 13.9% positive spots), which remained consistent with its stromal prioritization in Moffitt and its validation on the fibroblast side in USER. This does not imply exclusive expression in every fibroblast-containing spot, but it strongly supports a preferential association with stromal and CAF territories.

Taken these results together, *KCNN4* is better aligned with tumor epithelial territories, whereas *KCNMA1* is better aligned with stromal and CAF-rich territories. This spatial contrast completes the multi-level interpretation built from Moffitt, USER, and the local bulk perturbation model.

4.6. Cross-Dataset Limits and Interpretation of Low Ion Channel Expression

An important limitation of this project is that ion channels are often lowly expressed at transcript level across all the data types used here. This is true in scRNA-seq, where dropout and sparse counts reduce detection sensitivity; in snRNA-seq, where nuclear capture can underrepresent some transcripts; in bulk RNA-seq, where stromal signals are averaged across the whole sample; and in spatial transcriptomics, where spot-level averaging dilutes rare or localized signals. For this reason, transcript abundance alone is not a sufficient criterion to evaluate the relevance of ion channels in PDAC.

Another limitation is that many ion channel genes are not only weakly expressed, but can also have several transcript variants. Therefore, 3'-biased scRNA-seq, snRNA-seq, and spatial methods are useful for gene-level detection, but less informative for transcript-level complexity.

This limitation is (especially) important for potassium channels because their biological importance does not scale linearly with transcript abundance. Even a modest number of channels can shift membrane potential, calcium entry, cell volume regulation, or contractile behavior. In fibroblasts and myofibroblasts, these functions can directly affect matrix remodeling and tissue mechanics (Scruggs et al., 2020). In tumor cells, channels such as *KCNN4* can support calcium-dependent signaling and tumor progression (Jiang et al., 2019). In other words, low transcript abundance does not mean low functional significance.

This is why the workflow was built around convergence rather than single evidence. Instead of relying only on raw expression level, the project combined differential expression, patient-aware pseudobulk analysis, pathway association, perturbation response, and tissue localization. This does not remove all uncertainty, but it provides a more balanced way to study low-abundance but potentially high-impact genes.

5. Conclusions and Perspectives

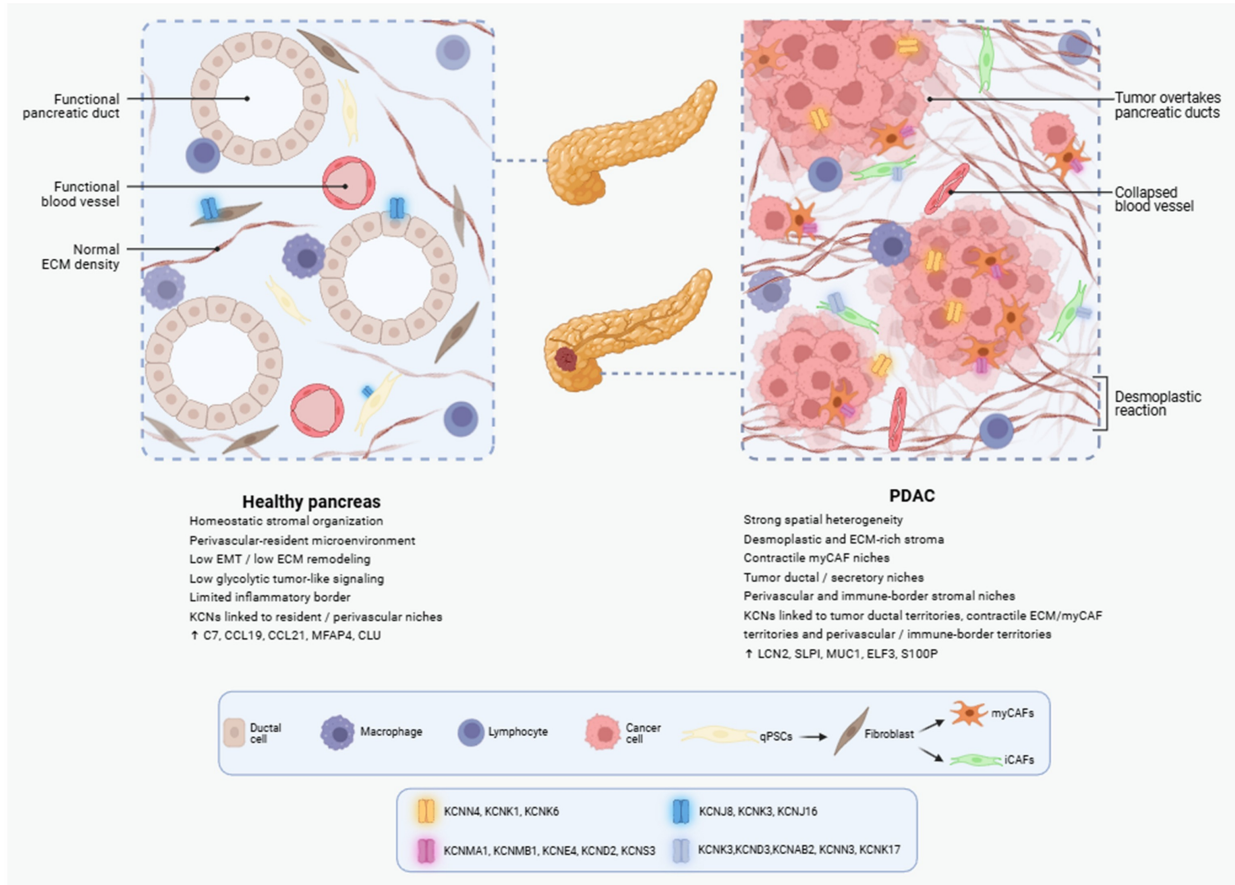


Figure 7: Summary of the selected KCN landscape in PDAC. The scheme summarizes the proposed relationship between PDAC stroma, tumor territories, and the main KCN candidates emphasized here, with a (simplified) comparison to non-diseased pancreas. The broader KCN union also suggests a directional organization of additional candidates across the PDAC microenvironment.

This report used a multi-level bioinformatic workflow to study potassium channel candidates in PDAC stroma. The main result is not the identification of one universal stromal channel, but the definition of a small and interpretable landscape. *KCNMA1* is the strongest candidate for an activated, matrix-rich, and contractile stromal pole. *KCNJ8* supports a more quiescent or resident stromal pole. *KCNN4* is the clearest tumor epithelial counterpoint and helps avoid an overly simplified all-stromal interpretation.

This model is coherent because it is supported by complementary data layers. The Moffitt stromal block provides the main stromal polarity. The USER atlas repositions the candidates in a broader tumor microenvironment. The local bulk RNA-seq model shows that BKCa silencing in CAFs is linked to measurable transcriptomic change. The spatial block shows that *KCNMA1* and *KCNN4* do not occupy the same biological territories in tissue.

The iCAF axis remains an important biological question, but it did not yield a KCN candidate as robust as *KCNMA1* or *KCNJ8* here. This should be interpreted as a limit of the current datasets and statistical power rather than as a proof that ion channels are irrelevant to iCAF

biology.

Future work can extend this study in four directions. First, the current workflow can be formalized (and will be) into a more reusable pipeline for additional ion channel families. Second, the strongest candidates can be tested experimentally in more direct stromal models to determine whether the BKCa axis centered on *KCNMA1* modulates fibroblast mechanics, matrix deposition, or stromal immune signaling. There is evidence from another fibrotic context, where the BK channel regulatory subunit *KCNMB1* has been linked to myofibroblast differentiation and pulmonary fibrosis (Scruggs et al., 2020). In another tumor microenvironment, recent single-cell and spatial analyses of brain metastasis identified a *KCNMA1*-high CAF subcluster within a contractile fibroblast compartment, giving additional support for a stromal and contractile interpretation of *KCNMA1* beyond PDAC (Simon et al., 2026). Together with findings linking BK signaling to myofibroblast differentiation in fibrosis (Scruggs et al., 2020), this suggests that *KCNMA1* may already be involved in stromal programs in other organs, while remaining relatively unexplored in PDAC. In this context, the results observed here may be consistent with an early indication of a similar stromal role in pancreatic cancer. The laboratory is currently exploring this direction on PDAC. Third, another direction I am currently exploring is MISTy (Multiview Intercellular SpaTial modeling), an interpretable machine-learning method for spatial data that helps separate intrinsic gene-expression patterns from neighborhood and tissue-context effects. I am interested in developing this analysis further to test whether *KCNMA1*, and more broadly the other KCN genes from the union, are mainly associated with stromal cell states or with specific multicellular spatial contexts in PDAC. Fourth, integrating spatial proteomics data would provide an important complement to the transcriptomic analyses presented here, as it would allow the spatial distribution of proteins to be assessed directly and help determine whether the observed transcriptional patterns are reflected at the functional protein level. This perspective is especially interesting because the broader KCN union already suggests a wider organization of potassium channel programs across the PDAC microenvironment.

Clinical Opening and Limitations

As a secondary extension, bulk survival analysis was used to ask whether the main transcriptomic candidates also leave a clinical signal at patient level. Among them, *KCNMA1* showed the clearest trend in the adjusted Cox model, with higher expression associated with better outcome (hazard ratio, HR = 0.80, meaning a lower risk; $p = 0.029$), although this signal did not remain significant after multiple-testing correction. By contrast, *KCNJ8* and *KCNN4* did not show a robust survival association across cohorts. This remains coherent with the rest of the report, because bulk survival averages stromal and tumor signals and cannot fully capture the interpretation supported by the single-cell and spatial analyses.

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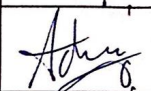
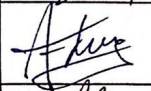
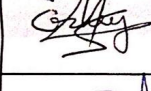
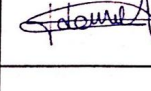
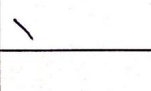
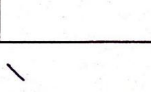
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M1 Skills Summary

During this internship, I learned how to organize a multi-level bioinformatic workflow, work with scRNA-seq, snRNA-seq, bulk RNA-seq, and spatial transcriptomic datasets, perform patient-aware pseudobulk analyses, and connect pathway enrichment results to an interpretable biological model. I also improved my ability to document analytical steps, generate readable figures, and discuss transcriptomic results with the level of caution required for low-abundance genes such as ion channels. More than just technical skills, this project showed me how exciting and complex cancer biology is when studied through bioinformatics. I learned that data analysis is about asking the right statistical questions, searching for the proper tools, and staying critical of the results. It requires a lot of trial and error, trying again and again even when things do not work or look blurry at first, and remaining calm to find the answers (at least, try to get some). So it taught me how to make "important" choices, justify them, and never stop questioning the data. I also improved my ability to communicate bioinformatic results with biologists, and to explain statistical or computational methods in simple terms so that they remained understandable and useful in discussion. Presenting my work during lab meetings also helped me gain confidence in speaking about data, defending analytical choices, and adapting explanations to different scientific backgrounds. Another important aspect of this internship was the motivation that comes from working on a concrete project in a laboratory environment. Seeing analyses become part of an active research question made the work much more meaningful and rewarding. It also showed me that many concepts learned during the Master's program, especially in omics and statistics, were directly useful in practice and really helped me move forward in this project. This internship reassured me that I made the right choice in following this path. It confirmed how much I enjoy bioinformatics, cancer-related questions, and molecular biology, showed me how much I still have to learn, and made me look forward even more to what comes next !

Seminar Attendance Sheet

Nom: <u>Mouroum</u>	Prénom: <u>Younna</u>	Master① ou 2: <u>B&C</u>
Nom du conférencier	Titre du séminaire	Nom et signature de l'hôte
Dr Yves Herveau Dr Gabriel Krouh Agathe Bayron Wendie Khouh 16/10/25	Conférence HODI	
Dr Danièle Raimondi Dr Sita Botton Suhail Khouh 16/10/25	Conférence HODI	
Marina Jule Augusto Amadio Kevin Labigand Pascal Benay 03/01/26	Conférence Xénium	
Hamoun Gueye 05/03/26	"Myosin 1D promotes de novo induction of zebrafish distal morphogenesis" IbV	
Julie Harvot 02/04/26	"Identification of novel targets of the TSC2 beta family member NODAL by RNA-seq" IbV	
Nils Ravel 16/09/26	Brain laterality in Drosophila: from sex to behaviour IbV	
Biocpu meeting Julien Heuer 29/09/26	"Analyzing the inner cells transcriptome up to down markers using scRNA-seq" Zoom	
Steno-se BGI Tech Jae li 04/05/26	Steno-seq: Spatial transcriptomics solutions IRCAN	
Biocpu meeting Jeanne Le Guez 3rd PhD Marilia Khouh 3rd PhD 22/05/26 22/05/26	Zoom Integrated Multiscale modeling of cellular ionic transport multiscale analysis of the pathophysiology of the fragile X syndrome	